B.TECH. COMPUTER SCIENCE AND ENGINEERING 2023

ESSENTIALS OF MACHINE LEARNING

Course Code:22CSE156 **Credits:**03
L:T:P:2:0:2:**2** **Contact Hours:** 60
Prerequisite(s): Python Programming

Course Objectives

The objective of the course is to

1. Introduce students to the basic concepts and techniques of Machine Learning
2. Have a thorough understanding of the Supervised and Unsupervised learning techniques
3. Study the various probabilities-based learning techniques
4. Study the techniques of Knowledge Representation and Intelligent Agents and to know the logical implications in computational intelligence.

Course Outcomes

At the end of the course, students will be able to

Course Outcomes	Description	Bloom'sTax onomyLevel
CO1	Understand the characteristics of Machine Learning models that are useful to solve real-world problems	Understanding(2)
CO2	Apply the different ML Models to create application.	Applying(3)
CO3	Apply appropriate ML algorithms for analyzing the data for variety of problems.	Applying(3)
CO4	Demonstrate different ML algorithms.	Applying(3)
CO5	Design the test procedures to assess the efficacy of the Developed model.	Applying(3)
CO6	Design and Combine several models to gain better results	Applying(3)

Course Contents

Module-1 **[12hours]**

INTRODUCTION TO ML:

Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches, Introduction to Numpy and Pandas, Datasets, Data Description in Pandas, Data Preprocessing, EDA and PDA

Module-2 **[12hours]**

REGRESSION AND CLASSIFICATION:

Linear Regression, OLS, MAE, SAME, RMSE, RSquare Classifications: Logistic Regression, Naïve Bayes, K-Nearest Neighbor, Confusion Matrix, and evaluation of the classification models.

SUPPORT VECTOR MACHINE:

Introduction, Types of support vector kernel–(Linear kernel, polynomial kernel, and Gaussian kernel), Hyperplane–(Decision surface), Properties of SVM, and Issues in SVM.

Module-3 **[12hours]**

DIMENSION REDUCTION & PARAMETER ESTIMATION:

Introduction and uses of Dimension Reduction and Parameter Estimation in ML, Hyper Parameter tuning, Principal Component Analysis (PCA), Linear Discriminate Analysis(LDA), Parameter Estimation: Maximum Likelihood Estimation

Module-4 **[12hours]**

DECISION TREE LEARNING–

Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning, Introduction to Random Forest

Module5: **[12hours]**

CLUSTERING & REINFORCMENT METHODS:

Introduction to Clusters, K-Means clustering, fixing the value of K in K-Means, Hierarchical Model, DBScan and Spiral Model

TEXTBOOKS

1. S.Russell and P.Norvig,"Artificial Intelligence: A Modern Approach ‖, Prentice Hall, Third Edition, 2009.
2. I.Bratko,―Prolog: Programming for Artificial Intelligence‖, Fourth edition, Addison-Wesley Educational Publishers Inc.,2011.
3. Bishop,C.,Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
4. M.Gopal, "Applied Machine Learning",McGraw Hill Education

REFERENCES

1. Stephen Marsland,― MachineLearning: An Algorithmic Perspective,CRCPress,2009.
2. Bishop,C.,Pattern Recognition and Machine Learning.Berlin:Springer-Verlag.
3. M.Gopal,"Applied Machine Learning", McGraw Hill Education
4. Tom M. Mitchell,―Machine Learning, McGraw- Hill Education (India) Private Limited, 2013.
5. Ethem Alpaydin,― Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press 2004.

Lab Course Contents:

Lab Exp#	Details
1	Train a Linear and Logistic Regression model to a given Data Set and Evaluate its performance
2	Train Different Classification Models to a Given Data Set evaluate its performance
3	Train Decision Tree/Random Forest Models to a Given Data Set evaluate itsperformance
4	Improve the machine learning model by tuning hyperparameters
5	Train different clustering model a Given Data Set evaluate its performance
6	Recommendation system from OTT/online content data using ML